Edward Silva

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Education

Colorado School of Mines May 2026

BS Electrical Engineering, Minor in Computer Science, Golden, CO, Graduate GPA: 4.00 (3 credit hours)

Honors: Dean's List (Fall 2022, Fall 2024, Spring 2026) | Honor Roll (Fall 2023, Spring 2024, Fall 2025)

Scholarships: Provost Scholarship | 2023–2024 C-MAPP Scholar (Frontier Technology Inc.) | 2024–2025 C-MAPP Scholar (Parsons Corporation) | 2024–2025 American Bureau of Shipping Scholar

Graduate: Estimation Theory & Kalman Filtering (A)

Relevant: Digital Signal Processing, Modern Control Design, Feedback Control Systems, Embedded Systems, Engineering Electromagnetics, Elements of Computing Systems, Printed Circuit Board Design

Experience

Embedded Software Engineer, SpaceX Jul 2026 – Present

C, C++, Git Bastrop, TX

- Write embedded C and C++ software and Linux kernel drivers for Starlink phased array systems
- Cut the run time of a calculation in the Starlink phased array software by 11%
- Interface with the RF, silicon, and network software teams on phased array software

Software Engineering Intern, Kratos Defense Jun – Aug 2025

C++, Python, SIMD, Git Colorado Springs, CO

- Sped up legacy real-time DSP code in C++ by 1.6x by refactoring and AVX-512 vectorization, guided by Intel VTune profiling
- Presented a faster FIR filter implementation and its performance trade-offs to the team for use on later projects
- Built a configurable logging framework with severity levels and error tracking for the real-time DSP software

Co-op Intern, Electrical Design, Jordan and Skala Engineers Jan – Jun 2025

Autodesk Revit, AutoCAD MEP, NEC Compliance Denver, CO

- Performed circuit loading and voltage drop calculations, sized breakers to NEC requirements, and drafted electrical layouts and design documentation for 20+ multi-unit residential and non-residential projects

Undergraduate Researcher, ePower Hubs Research Lab, Golden, CO Aug – Dec 2024

- Reviewed the literature on sensor systems and farm-level control strategies for offshore wind, under variable voltage, power, and frequency constraints
- Wrote the findings up in internal LaTeX reports on cost reduction in wind farm maintenance, design, and grid integration

Calculus II and Differential Equations Grader, Colorado School of Mines, Remote Spring – Summer 2024

- Graded Calculus II in Spring 2024 and Differential Equations in Spring and Summer 2024

Peer Mentor, Colorado School of Mines, Golden, CO Aug – Dec 2023

- Co-led a one-credit course that helped first-year students settle in, and led a group of new students through orientation

Projects

Electromagnetic Bicycle Damper (EMBR), MATLAB, Python Fall 2025 – Spring 2026

- Owned the electromagnet design and calculation verification within an 8-person interdisciplinary team, sponsored by ARUS MR Tech, that retrofitted a production front suspension fork with a semi-active MR damper
- Modeled a ∼250-turn, 2 A coil in Python by summing Biot-Savart loop fields with complete elliptic integrals, estimating a 0.15 T peak across the 2.5 mm radial gap

- Helped a teammate design the Arduino PWM control that varies the coil current, and with it the magnetic field

Nonlinear State Estimation for the EMBR Damper (graduate course, solo), MATLAB, Simulink Spring 2026

- Modeled the MR damper, MOSFET-driven coil, and sensor array as a 4-state nonlinear system in MATLAB and Simulink, adding terrain force to the state as a random walk
- Implemented and compared Extended (EKF) and Unscented (UKF) Kalman Filters to estimate unmeasured states
- At a larger yield constant, the UKF cut the RMS force error from 624 N to 401 N and kept every sample inside its 3-sigma bounds

Pitch-Tracking Systems in Auto-Tune, MATLAB Spring 2026

- Compared real-time FFT and autocorrelation (ACF) pitch trackers in MATLAB, finding in a 50 Hz to 4 kHz chirp sweep that the ACF failed near 1600 Hz and the FFT near 1800 Hz
- Measured mean per-frame latency of 0.22 ms for the FFT against 0.59 ms for the ACF (2.68x), with 0.26 ms against 0.74 ms standard deviation

Gantry Crane Control System, MATLAB, Simulink Fall 2025

- Designed a 50 Hz discrete-time gantry-crane controller in the z-domain with 0% overshoot, 112.6 s settling against a 120 s spec, and cart velocity ≤ 1 m/s
- Found a 48° phase margin and 0.52 s delay tolerance, confirmed by marginal instability with an injected 25-sample delay
- Pair-programmed an LQR state-space controller with a full-order observer placed about 6x faster than the controller
- Compared reference scaling (3.95 s settling, 9.02% overshoot) with integral augmentation plus anti-windup (3.45 s, negligible overshoot)

Autonomous Maze-Navigating Robot, Python Fall 2025

- Achieved the second-fastest course completion time in the cohort with an autonomous maze-navigating robot built by a 4-person team, integrating computer vision, embedded control, and I2C communication
- Built the computer vision subsystem with one teammate, using Python and OpenCV with ArUco marker detection to extract real-time distance and angle metrics for navigation
- Split the 640x480 vision pipeline into capture, processing, display, and LCD threads so the Arduino stopped waiting on the Python
- Fixed an I2C packet mismatch by repacking 8-byte Python doubles as 4-byte little-endian floats for the Arduino reader

Hydraulic Ram Pump Fall 2022

- Built a sub-$100 PVC ram pump prototype in a 5-person team whose filter raised test water from pH 5 to pH 9 (pool test strips), and received the Subject Matter Expert Award

eeprep, GitHub Ongoing

- Wrote eeprep, a free guide for EECS students and new graduates looking for a first job

Skills

Languages: C, C++, Python, MATLAB
Tools: Simulink, Git, Intel VTune, OpenCV, Arduino, Autodesk Revit, AutoCAD MEP
Methods: Kalman filtering (EKF, UKF), LQR and observer design, SIMD (AVX-512), FIR filters, I2C

Certifications

MATLAB Machine Learning Techniques, MathWorks

Organizations

IEEE-HKN, Nu Pi Chapter, Colorado School of Mines, Member 2026 – Present
IEEE, Member 2025 – Present
IEEE Young Professionals, Member 2025 – Present